

CreditFair

A Fair, Explainable, Human-Supervised Credit Scoring System

Project Report

Responsible AI & Data Ethics Project — SRH Heidelberg

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Dataset: UCI Statlog German Credit Data (1,000 applicants)

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Overview

CreditFair is a responsible credit-scoring system: a model that predicts whether a loan applicant is likely to repay, designed from the ground up to be fair across groups, explainable to the people it affects, and always checked by a human before a final decision is made. The project uses a well-known public dataset of 1,000 historical loan applications (the UCI “German Credit” dataset) and works through the entire lifecycle of building such a system responsibly: understanding the data and its biases, building and comparing prediction models, measuring and fixing unfairness, explaining decisions in plain language, stress-testing the system, and packaging everything into a working, tested, documented product.

The finished product has four parts: a tested scoring engine (`src/creditfair.py`) that turns an applicant's information into a risk score, a fair decision, and a plain-language explanation; a suite of 31 automated tests that catch defects and fairness regressions before they reach production; an interactive dashboard that lets a credit officer review, explain, and act on decisions; and a governance package (model card, data-protection and fundamental-rights impact assessments, and a fairness-to-law mapping) that documents the system the way a real deployer would be required to.

The headline result: without any correction, the model's raw predictions treat some groups (younger applicants, foreign workers) noticeably worse than their actual repayment behavior justifies. After applying a fairness-aware training method, approval-rate gaps for sex and foreign-worker status are brought above the standard 80% fairness threshold, and the age gap is brought to the edge of it at a modest, clearly reported cost. Every rejection is automatically sent for human review, and the system explains its reasoning in plain language for every applicant.

1. Project Purpose and Goals

Banks and lenders increasingly use automated models to decide whether to approve a loan. Left unchecked, these models can unintentionally treat people unfairly based on characteristics like sex, age, or nationality even when those characteristics are never directly used because other information in the application (marital status, housing, job type) can act as a stand-in, or “proxy,” for the protected trait. CreditFair was built to show, concretely, how to build a credit-scoring system that avoids this trap while still being accurate and useful.

The project set out to achieve four goals:

- **Goal 1 — Explainability:** make every risk score and every decision understandable, to the credit officer reviewing it.
- **Goal 2 — Robustness:** test that the system behaves sensibly under noisy inputs, shifting populations, and the known weaknesses of historical lending data.
- **Goal 3 — Testing:** back the system with an automated test suite that would catch a broken, unfair, or regressed model before it ships.
- **Goal 4 — A single, deployable decision path:** package the whole pipeline, risk scoring, fairness correction, explanation, and human-review routing behind one clean interface that a real application could call.

Underlying all four goals is one design principle that shapes every decision in the project: the model may advise, but a human always decides. It is enforced by a review-routing rule that sends every rejection, and a sample of other borderline or higher-risk cases, to a person before anything is finalized.

2. The Dataset

The project uses the Statlog German Credit dataset, a well-known public dataset from the UCI Machine Learning Repository, originally compiled by Prof. Hans Hofmann in 1994. It contains 1,000 real historical loan applications, each described by 20 attributes (financial history, employment, housing, loan purpose, and demographic information), together with the outcome the bank observed: whether the applicant turned out to be a “good” or “bad” credit risk.

- 1,000 applicants in total: about 700 labelled “good” risk and 300 labelled “bad” risk (a 70/30 split, moderately imbalanced).
- No missing values, the dataset uses explicit categories such as “no checking account” rather than blank fields, so the absence of an account is itself meaningful information.
- Three attributes are legally sensitive (“protected”) and were tracked specifically for fairness auditing: sex, age, and foreign-worker status. Foreign-worker status is treated as a stand-in for ethnic origin, a recognized protected category.

A defining feature of this dataset and of real-world lending decisions is that a wrong decision is not equally costly in both directions. Approving a loan that is never repaid is far more expensive than declining a loan that would have been repaid. The dataset's official cost matrix reflects this: approving a bad applicant costs five times as much as rejecting a good one (a 5:1 ratio). This 5:1 cost ratio is used throughout the project to judge every model, every decision threshold, and every fairness of trade-off, not plain accuracy, which would hide this asymmetry.

	Predicted Good	Predicted Bad
Actually Good	0 (correct)	1 (cost of an unnecessary refusal)
Actually Bad	5 (cost of an unpaid loan)	0 (correct)

Table 1. The official 5:1 cost matrix: approving a bad-risk applicant costs five times as much as wrongly refusing a good one.

3. Development Process

The project was carried out in stages of work, captured in a single merged notebook covering data analysis, model building, and final testing and explainability. The sections below combine and simplify that work into a logical order, removing technical jargon.

3.1 Understanding the Data

The first step was to load the raw data file, decode its coded values into readable categories (for example, translating internal codes into labels like “critical credit history”), and checking its quality. The data quality check confirmed there were no missing values and that the 30% bad-risk rate was significant enough to require careful evaluation (plain accuracy would be misleading, since a model that approved everyone would still look 70% “accurate”).

A first look at the data's relationships surfaced a few clear patterns:

- Loan duration is the strongest numeric predictor of risk; longer loans carry more risk followed by loan amount; older applicants tend to be slightly lower risk.
- An applicant's checking-account status is the single strongest predictor overall: applicants with a negative balance are bad-risk about 49% of the time, versus about 12% for applicants with no account at all.

- Credit history shows a counter-intuitive pattern: applicants with a “critical” credit history (i.e., some past problems) actually looked lower-risk (~17%) than applicants with no credit history at all (~63%) — a reminder that a track record, even an imperfect one, carries information.

3.2 Measuring Fairness in the Raw Data

Before building any model, the team measured how the historical outcomes themselves differed across the three protected groups sex, age, and foreign-worker status to understand what biases a model trained on this data would risk inheriting.

- Sex: women were historically labelled bad-risk 35% of the time versus 28% for men.
- Age: the youngest applicants (18–25) were labelled bad-risk 42% of the time, versus 24% for the 36–50 group in the widest age gap in the data.
- Foreign workers were labelled bad-risk 31% of the time versus 11% for non-foreign applicants. A large gap, though based on only 37 non-foreign applicants, makes the estimate fragile.

The team also checked whether simply removing sex and foreign-worker status from the data would be enough to prevent the model from discriminating. It found that other, related features (job type, housing, marital status) could still predict sex with about 70% accuracy meaning the model could reconstruct a protected trait indirectly, even without seeing it directly. This “proxy” problem is a central theme that shapes the rest of the project's fairness work.

3.3 Preparing the Data for Modelling

The cleaned data was split into three parts, in fixed proportions and using the same random seed for reproducibility: 600 applicants to train the models, 200 to validate and tune them, and 200 held back and used only once, at the very end, to report final performance. This separation is standard for good practice as it prevents the model from being unintentionally tuned to the exact data it will be judged on.

A small number of applicants had unusually long loan durations or unusually large loan amounts, which could distort the model. These extreme values were capped at the 99th percentile observed in the training data only, then the same cap was applied to the validation and test data. Calculating this cap strictly from the training set (rather than from the whole dataset) is a leakage-prevention step: it ensures the validation and test sets stay genuinely unseen, as they would be in production. Applicant age was left uncapped since it did not show extreme outliers.

Sex and foreign-worker status were deliberately excluded from the information given to the model at decision time using a protected characteristic to decide is direct discrimination and is not permitted. Age, a lawful and genuinely predictive factor in consumer credit, was kept. The excluded attributes were retained separately, purely for auditing fairness afterward, never for scoring.

3.4 Building and Comparing Models

Three candidate models were trained and compared to the validation data: Logistic Regression (a simple, transparent baseline), Random Forest, and Gradient Boosting (two more powerful “ensemble” models). Because a false approval is five times costlier than a false rejection, each model's decision threshold — the risk score above which an application is rejected was tuned separately for each model to minimize the 5:1 cost on the validation data, rather than using a generic 50% cutoff. Every model's optimal threshold turned out to be well below 50%, reflecting how costly it is to miss a genuinely bad-risk applicant.

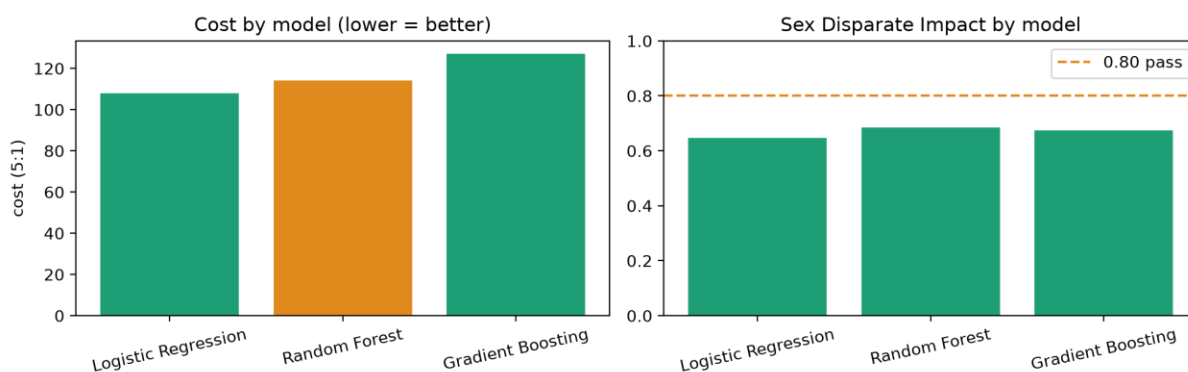


Figure 1. Left: The total cost (lower is better) for each candidate model. Right: how fairly each model treats sex, before any correction — all three fall well short of the 80% fairness benchmark.

Random Forest was chosen as the working risk-scoring model. It was not the cheapest option in isolation, but the comparison made clear that the cheapest model is not automatically the fairest one cost, and fairness had to be weighed together, not cost alone. On the held-out test set, the chosen model reached a ROC-AUC of approximately 0.79 (a measure of how well it ranks risky applicants above safe ones, where 0.50 is a coin flip and 1.00 is perfect), with validation and test performance close enough to confirm the model generalizes rather than memorizing the training data.

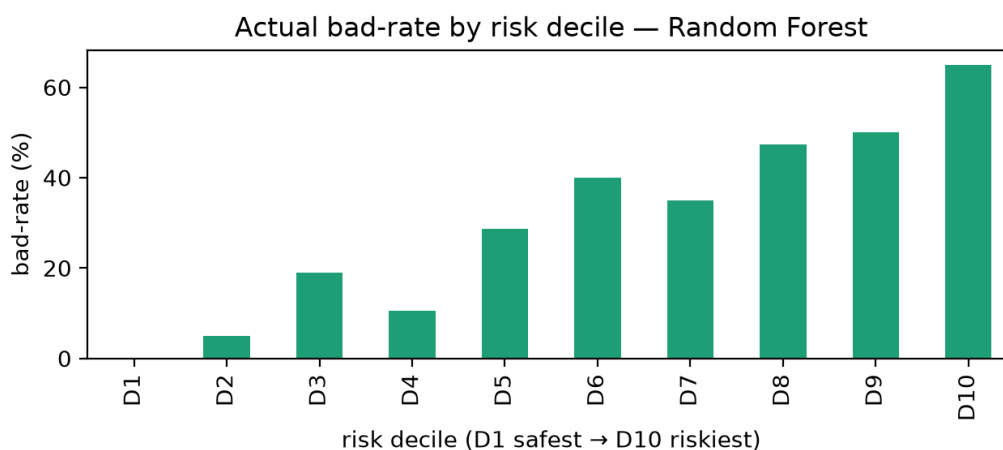


Figure 2. Applicants sorted from safest (D1) to riskiest (D10) decile by model score. The steadily rising actual bad rate confirms the score is a genuinely useful risk ranking.

3.5 Measuring Fairness of the Model's Decisions

With a working model chosen, the team measured how fairly its decisions treated each protected group, using the standard “four-fifths rule” (also called disparate impact): if the least-favored group's approval rate is below 80% of the most-favoured group's rate, the outcome is flagged as a fairness concern.

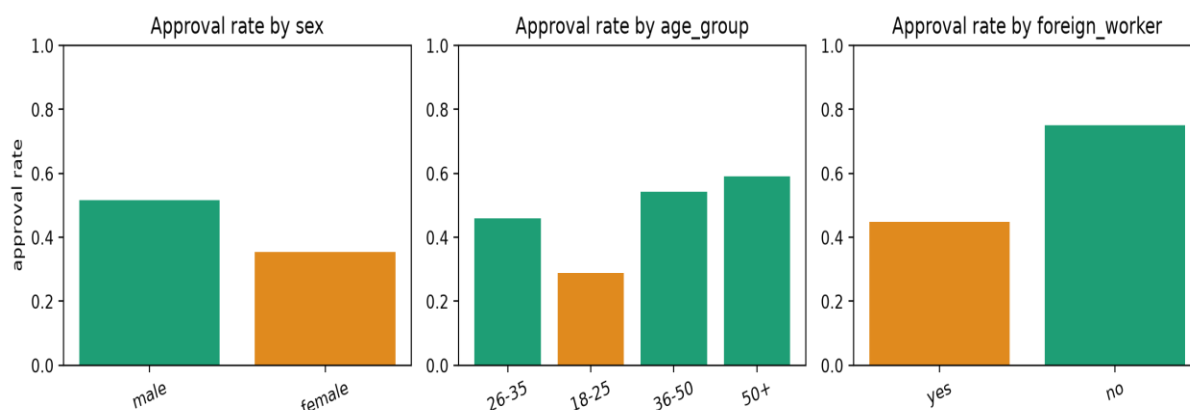


Figure 3. Baseline approval rates by group. Women, the youngest applicants, and foreign workers are approved less often than their counterparts.

Before any correction, all three protected attributes fell short of the 80% fairness line: sex scored 0.84 (borderline pass), while age (0.56) and foreign-worker status (0.61) clearly failed. Because the test set is only 200 rows, the team also calculated a 95% confidence interval around each figure using bootstrap resampling and reported those honestly rather than quoting a single precise-looking number the intervals were wide, especially for the smaller subgroups.

Two further checks went beyond the headline number. First, an “amplification” check compared the model's approval-rate gap for each group against the gap that group's actual repayment behavior would justify finding that the model exaggerated the historical gap for all three attributes (for example, foreign workers' true risk profile would justify roughly a 0.79 approval ratio, but the model delivered only 0.61). Second, a calibration check found the model's predicted risk was not simply over-estimating risk for any one group the disparity was not a straightforward technical bug, which meant it needed an active fairness correction rather than a simple recalibration fix.

3.6 Why Just Removing Protected Attributes Is Not Enough

Sex and foreign-worker status were never given to the model as inputs — but the team tested directly whether that alone was sufficient to prevent discrimination, using two checks. A “proxy audit” trained a small separate model to predict an applicant's sex purely from the remaining, “lawful” features, and found it could do so far better than chance confirming that features like job, housing, and marital status carry a meaningful trace of the protected trait. A “counterfactual” check then asked: does an applicant's decision change if a sex-linked feature (marital status) is swapped, holding everything else fixed? On this project's data split, swapping marital status barely moved the fairness numbers, suggesting the residual bias for sex was spread diffusely across many features rather than concentrated in one obvious proxy.

The conclusion drawn was an important one for the rest of the project: removing a protected attribute from the model's inputs is a necessary legal safeguard against direct discrimination, but it is not, by itself, a fix for fairness. An active correction step was still required.

3.7 Correcting for Unfairness

Different fairness-correction techniques were built and compared on the same fairness-versus-cost basis, for every protected attribute:

- **Pre-processing (reweighing):** adjust the training data, so outcomes and group membership become less correlated, before training. This was tested and found to make fairness worse, not better, on this data and was rejected.
- **In-processing (Exponentiated Gradient):** constrain the model during training to equalize approval rates across sex, age, and foreign-worker status jointly, while never using those attributes to make the actual decision. This was the strongest performer and was adopted.
- **Post-processing (per-group thresholds):** apply a different approval cutoff to each group after the fact. This scored well numerically but was rejected on legal grounds setting up a different bar for different groups to use the protected trait directly at decision time, which is itself a form of direct discrimination.

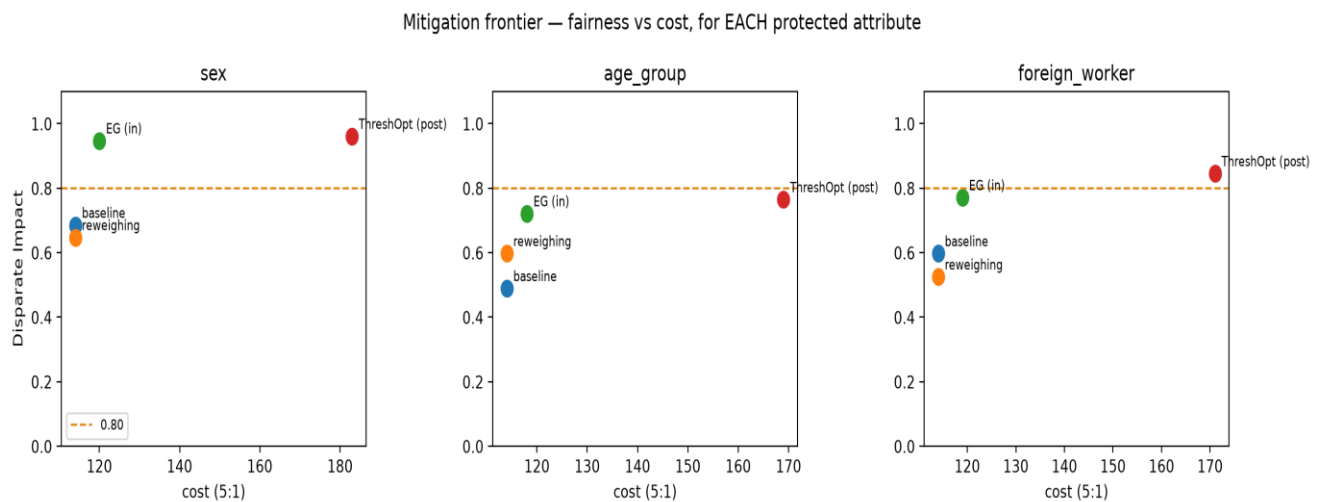


Figure 4. Fairness (vertical axis) versus cost (horizontal axis) for each correction method, per protected attribute. The in-processing method (green) reaches or nears the 0.80 fairness line at low added cost, without resorting to the legally risky post-processing approach (red).

The adopted method is an intersectional, cost-aware Exponentiated Gradient correction that constrains the model to treat combinations of sex, age, and foreign-worker status fairly together (not just one attribute at a time), while staying tuned to the same 5:1 cost structure used throughout the project.

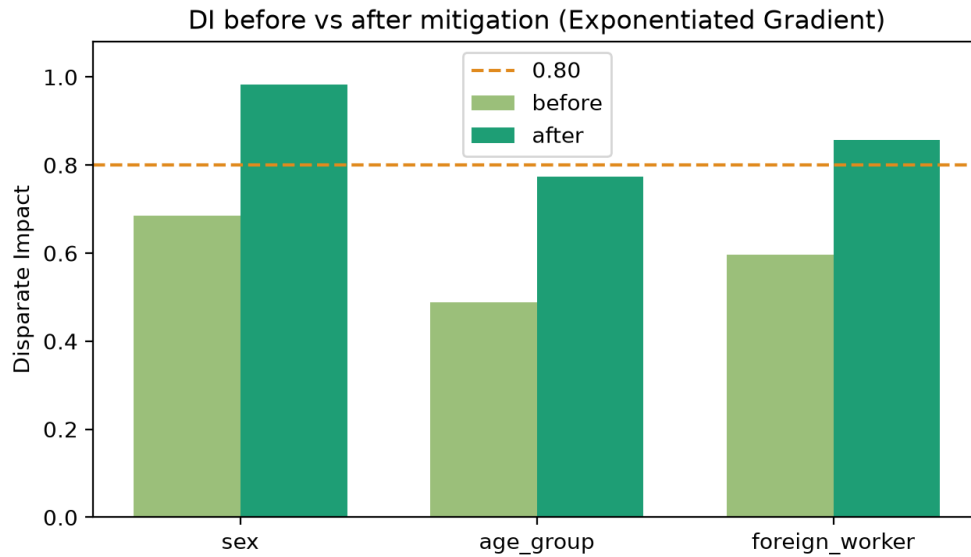


Figure 5. Fairness (disparate impact) for each protected attribute before and after the correction. Sex and foreign-worker status clear the 0.80 line; age reaches it closely but remains borderline.

Attribute	Before correction	After correction	Result
Sex	0.68	~0.95	Clears the 0.80 fairness line
Foreign-worker status	0.59	~0.87	Clears the 0.80 fairness line
Age	0.48	~0.77	Reaches the line; borderline (varies slightly by machine)

Table 2. Fairness results before and after correction. The correction added a modest cost (roughly +25 on the project's cost scale, moving expected cost from about 115 to about 140) — the documented, deliberate price of fairness.

The team was candid about a trade-off inherent to this approach: equalizing approval rates across groups (“demographic parity”) narrows the approval-rate gap but can slightly widen the gap in error rates (how often genuinely good or bad applicants are misclassified). This was measured and reported honestly rather than hidden, alongside a note that age fairness, while improved, remains close enough to the line that it should not be declared fully “solved” it is instead backstopped by mandatory human review.

3.8 Explaining Every Decision

A model that cannot explain itself cannot be meaningfully reviewed by a human or meaningfully contested by an applicant. The project builds explanations at two levels.

First, reason codes: an interpretable companion model (Logistic Regression) was used to produce a short, ranked list of the features pushing a given decision toward approval or rejection, giving a human reviewer an immediate, plain read on why a case scored the way it did.

Second, deeper explainability using SHAP (SHapley Additive exPlanations), a well-established technique for explaining what a complex model has learned. Applied to the Random Forest risk score, it confirmed the model had learned economically sensible signals rather than noise.

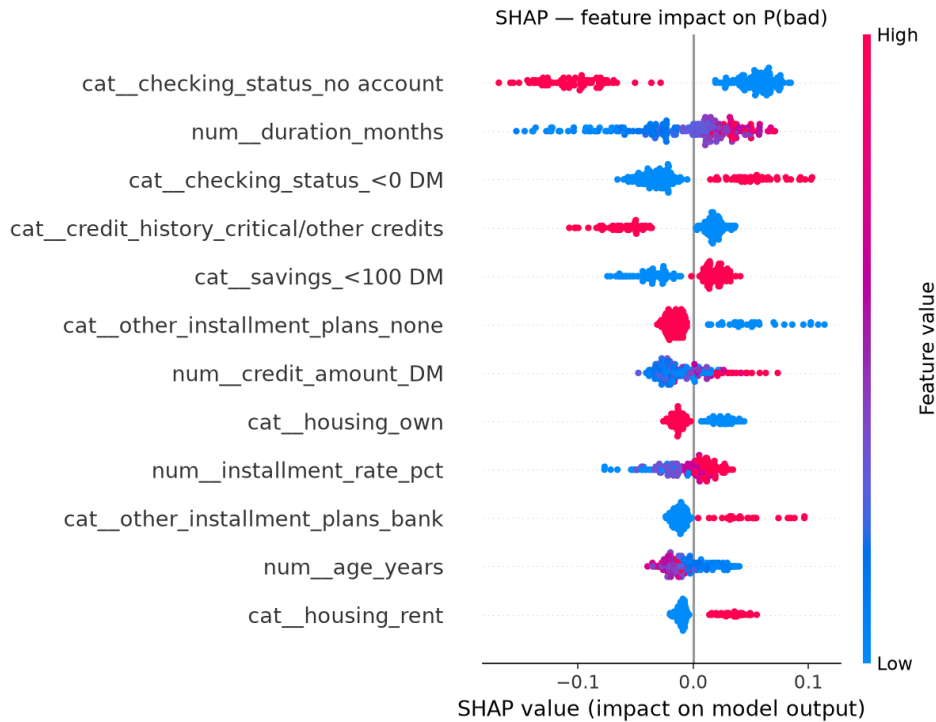


Figure 6. SHAP summary: each dot is a test applicant. Points to the right push risk higher. No or negative checking account balance, longer loan duration, low savings, and a weak credit history all push risk upward — consistent with real-world lending intuition.

3.9 Stress-Testing the System

Beyond accuracy and fairness on a single test run, the project checked whether the system holds up under more realistic, imperfect conditions:

- **Input noise:** small random perturbations were added to applicant data to see whether decisions flipped unpredictably. Decisions were found to be stable under small noise.
- **Subgroup uncertainty:** the model's confidence was examined specifically for smaller, historically disadvantaged subgroups, where fewer training examples mean more uncertainty.
- **Population drift:** a monitor (the Population Stability Index, or PSI) was built to detect if a future batch of applicants looks meaningfully different from the population the model was trained on — for example, a shift toward longer, larger loans. The monitor was shown to correctly stay quiet on an in-distribution sample and fire on a simulated shifted one.

3.10 Automated Testing

The full pipeline was backed by an automated test suite of 31 tests, covering approximately 99% of the scoring engine code, run with the pytest framework. A live run of the suite, executed as the final step of the notebook, confirmed all 31 tests passing. The tests fall into two categories.

Most tests check straightforward correctness: that the data loads and decodes properly, that the train/validation/test split has no leakage between the three parts, that the 5:1 cost calculation is arithmetically correct, that every applicant is routed to exactly one human-review priority tier, and that the deployed decision-making function returns sensible, complete output.

Six tests are described as “weakness-detectors”, tests specifically designed to fail if the system were quietly broken in a way that ordinary tests might miss. These include a test that re-trains the fairness correction and fails the whole build if it stops improving sex fairness; a test that constructs a maximally risky and a maximally safe hypothetical applicant and checks the risky one always scores higher (which would catch a model whose predictions are accidentally inverted); a test confirming that making one input strictly worse (e.g., lower savings) never lowers an applicant's calculated risk; and tests confirming that sex, foreign-worker status, and related identifying information can never leak back into the model's decision inputs.

Test category	Number of tests	What it protects against
Data integrity	3	Corrupted or mis-decoded input data
Split & leakage	4	Test data accidentally influencing training
Cost & fairness metric math	5	Incorrect cost or fairness calculations
Human-review routing	5	An applicant falling through without review
Reweighting & reproducibility	2	Inconsistent results between runs
Model behavior (real fit)	2	Broken fairness summaries or reason codes
Deployable model (single decide() call)	4	A broken end-to-end decision path
Weakness-detectors	6	Fairness regressing, an inverted model, proxy leakage, overfitting

Table 3. The 31-test automated suite, grouped by what each group of tests protects against.

4. Final Deliverables

CreditFair is a complete, working product, and it consists of four pieces connected.

4.1 The Scoring Engine (src/creditfair.py)

All the tested logic, data loading, the Random Forest risk scorer, the fairness-corrected decision model, reason-code generation, and human-review routing, live in a single, reusable Python module. It exposes one clean entry point, `CreditFairModel.decide()`, which takes an applicant's information and returns a risk score, a decision, a review priority, and a plain-language set of reasons in one call, directly fulfilling the goal of a single, deployable decision path.

4.2 The Human-Review Rulebook

Every single decision the system makes is routed to a human reviewer, sorted into one of four priority tiers, so that oversight effort is concentrated where it matters most:

- **P1 — every rejection:** mandatory review, satisfying the legal expectation that adverse automated decisions are checked by a person.

- **P2 — borderline approvals:** approvals that sit close to the decision cutoff.
- **P3 — approvals in disadvantaged subgroups:** extra scrutiny for the groups in the fairness analysis flagged as most exposed.
- **P4 — clear approvals:** reviewed on a sampled basis, since these are the lowest-risk, highest-confidence cases.

4.3 The Dashboard

A Streamlit-based interactive dashboard gives a credit officer a working interface to the whole system, organized into seven tabs:

- Overview — headline performance and fairness numbers, plus the SHAP explanation of what drives risk.
- Fairness monitor — live disparate impact tracking across the protected groups.
- Drift monitor — an interactive view of the Population Stability Index, flagging when incoming applicants look different from the training population.
- Review queue — where a human reviewer confirms or overrides individual recommendations, with bulk approval for the lowest-risk (P4) tier.
- Credit assessment — scores a new applicant and shows the verdict alongside a plain-language summary.
- Audit log — a session record of decisions and reviewer actions, exportable to CSV.
- Compliance — a summary view tying the system's behavior back to its governance documentation.

4.4 Governance and Compliance Documentation

Because credit scoring is a high-stakes, regulated use of AI, the project produced the documentation a real deployer would be expected to maintain, written by the project team (Simran Amesar and Shreya Challagulla):

- **Model Card:** a single technical summary of the model — what it is, how it was trained, its intended use and its explicit out-of-scope uses, its performance and fairness numbers, and its known limitations.
- **Data Protection Impact Assessment (DPIA):** a structured risk assessment (required under GDPR Article 35 for this kind of automated decision-making) covering what data is processed, the risks to applicants, and the mitigations in place for each one.
- **Fundamental Rights Impact Assessment (FRIA):** an assessment under the EU AI Act (Article 27, applicable to high-risk credit-scoring systems from 2 August 2026) of the system's impact on rights such as non-discrimination and the right to an explanation.
- **Fairness-to-Law Map:** a reference document connecting each fairness technique used in the project (removing protected attributes, measuring disparate impact, the adopted correction method, human review, and so on) to the specific legal requirement it addresses, and honestly noting what each technique does and does not achieve on its own.

5. Key Outcomes

Bringing together the results from across the development process:

Area	Result
Model performance	ROC-AUC ≈ 0.79 on held-out test data; validation and test performance are close, indicating the model generalizes rather than memorizing.
Cost (5:1 scale)	Baseline expected cost ≈ 115 ; after fairness correction ≈ 140 — a modest, transparently reported “price of fairness.”
Fairness — sex	0.84 before correction $\rightarrow \approx 0.95$ after (clears the 0.80 fairness line).
Fairness — foreign-worker status	0.61 before $\rightarrow \approx 0.87$ after (clears the 0.80 fairness line).
Fairness — age	0.56 before $\rightarrow \approx 0.77$ –0.84 after (reaches the line; treated as borderline, not solved).
Human oversight	100% of decisions routed to a human reviewer; every rejection is mandatory review (P1).
Explainability	Every decision carry reason code and a plain-language adverse-action notice guaranteed non-empty for rejections.
Automated testing	31 tests, ~99% code coverage, all passing on a live run, including 6 dedicated “weakness-detector” tests.

6. Limitations

The project documents its own limitations candidly rather than overstating its results:

- **Small test set:** with only 200 held-out applicants, fairness figures carry wide statistical uncertainty (shown via bootstrap confidence intervals), and some numbers shift slightly (± 0.05 – 0.08) depending on the machine they are run on, though the overall pattern is stable.
- **Selection bias in the historical data:** outcomes are only known for applicants a 1990s bank actually approved, which can distort both performance and fairness estimates. This was quantified but not fully corrected.
- **Age fairness is borderline:** the correction brings age close to the 0.80 fairness line but not decisively above it on every run — addressed through mandatory human review rather than declared fully resolved.
- **Dated data:** the underlying dataset reflects 1990s German lending and may not represent a current applicant population without re-validation.
- **Scope of explainability:** the SHAP explanations describe the underlying risk score; the fairness-corrected decision itself is explained separately, using a method suited to that model structure.

7. Conclusion

CreditFair demonstrates, end to end, what it takes to build a credit-scoring system responsibly: understanding a dataset's built-in historical biases, proving that simply hiding protected attributes is not enough, measuring fairness honestly (including its statistical uncertainty), choosing a correction method that is both effective and legally sound, explaining every decision in plain language, keeping a human in charge for model outcomes, and backing all of it with automated tests that would catch a regression before it reached a real applicant.

The result is not a system that claims to be perfectly fair it is one that measures its own shortcomings precisely, corrects what it reasonably can, and is explicit and honest about what remains borderline or unresolved, while ensuring no applicant is ever left to a fully automated decision. That combination of technical rigor and transparent honesty is the project's core achievement.